

STUDENT PERFORMANCE ANALYSIS USING EXPLORATORY DATA ANALYSIS (EDA)

Comprehensive Project Report

Prepared By: Vansh Bathla

Data Analytics Internship Project

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

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Executive Summary

This project investigates the factors affecting student academic performance using exploratory data analysis. The dataset contains demographic, social, family, and academic attributes for students. The objective is to identify patterns influencing final grades and derive actionable insights. Multiple visualizations, descriptive statistics, and correlation techniques were used to understand the relationships between study habits, previous grades, failures, attendance, and overall academic outcomes. The findings highlight that previous academic performance and study time are strong indicators of future success while failures negatively impact performance.

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Dataset Overview

Rows: 395

Columns: 33

Missing Values: 0

The dataset includes variables related to demographics, parental education, lifestyle factors, study habits, attendance, and grades.

Feature Dictionary

Feature	Description
school	School
sex	Gender
age	Age
address	Address type
famsize	Family size
Pstatus	Parent status
Medu	Mother education
Fedu	Father education
Mjob	Mother job
Fjob	Father job
reason	School choice reason
guardian	Guardian
traveltime	Travel time
studytime	Study time
failures	Past failures
schoolsup	School support
famsup	Family support
paid	Extra classes
activities	Activities
nursery	Nursery school
higher	Higher education
internet	Internet access
romantic	Relationship status
famrel	Family relationship
freetime	Free time
goout	Going out
Dalc	Workday alcohol
Walc	Weekend alcohol
health	Health
absences	Absences
G1	Grade 1
G2	Grade 2
G3	Final Grade

Data Quality Assessment

Data quality checks were performed to identify missing values, inconsistencies, and structural issues. The dataset contains no missing values and is suitable for direct analysis. Numeric and categorical features were validated before generating statistical summaries and visualizations. Clean and reliable data improves the credibility of analytical conclusions.

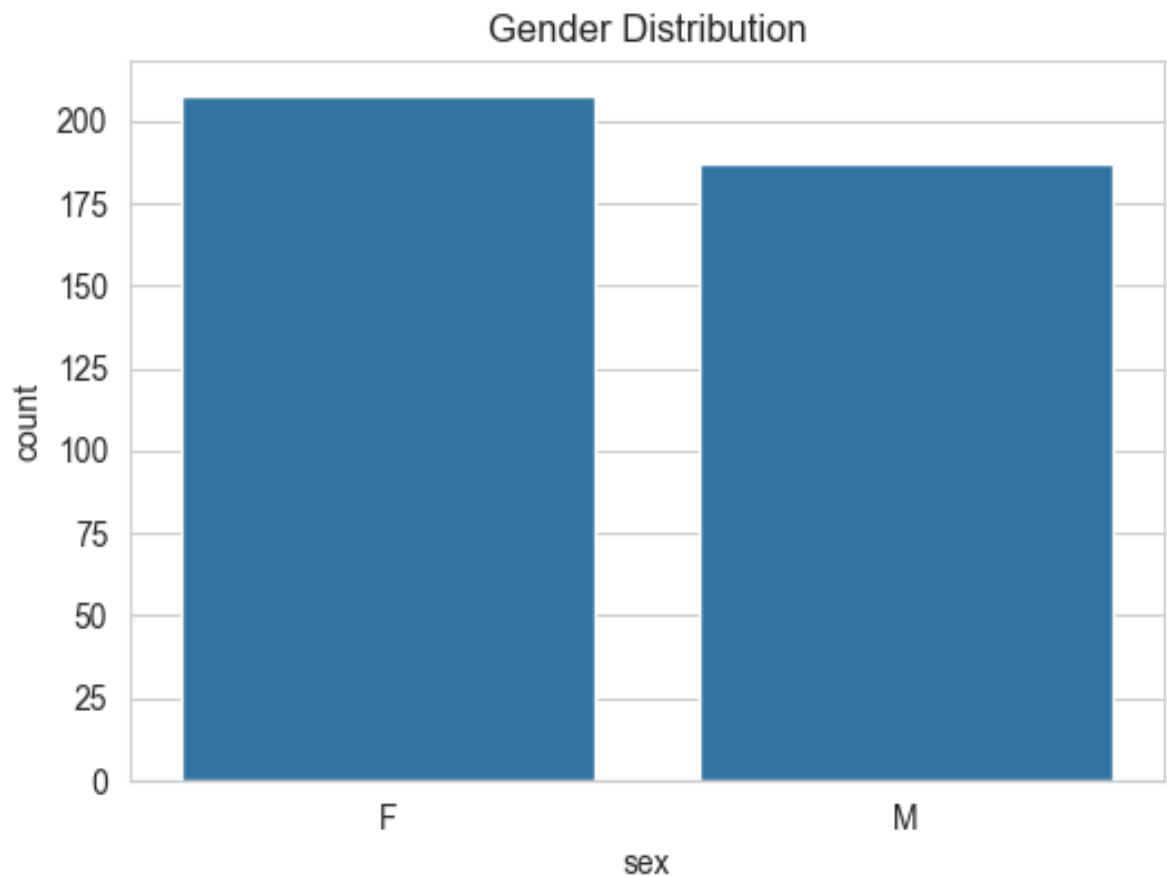
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Statistical Overview

Metric	age	studytime	failures	absences	G1	G2	G3
count	395.0	395.0	395.0	395.0	395.0	395.0	395.0
mean	16.7	2.04	0.33	5.71	10.91	10.71	10.42
std	1.28	0.84	0.74	8.0	3.32	3.76	4.58
min	15.0	1.0	0.0	0.0	3.0	0.0	0.0
25%	16.0	1.0	0.0	0.0	8.0	9.0	8.0
50%	17.0	2.0	0.0	4.0	11.0	11.0	11.0
75%	18.0	2.0	0.0	8.0	13.0	13.0	14.0
max	22.0	4.0	3.0	75.0	19.0	19.0	20.0

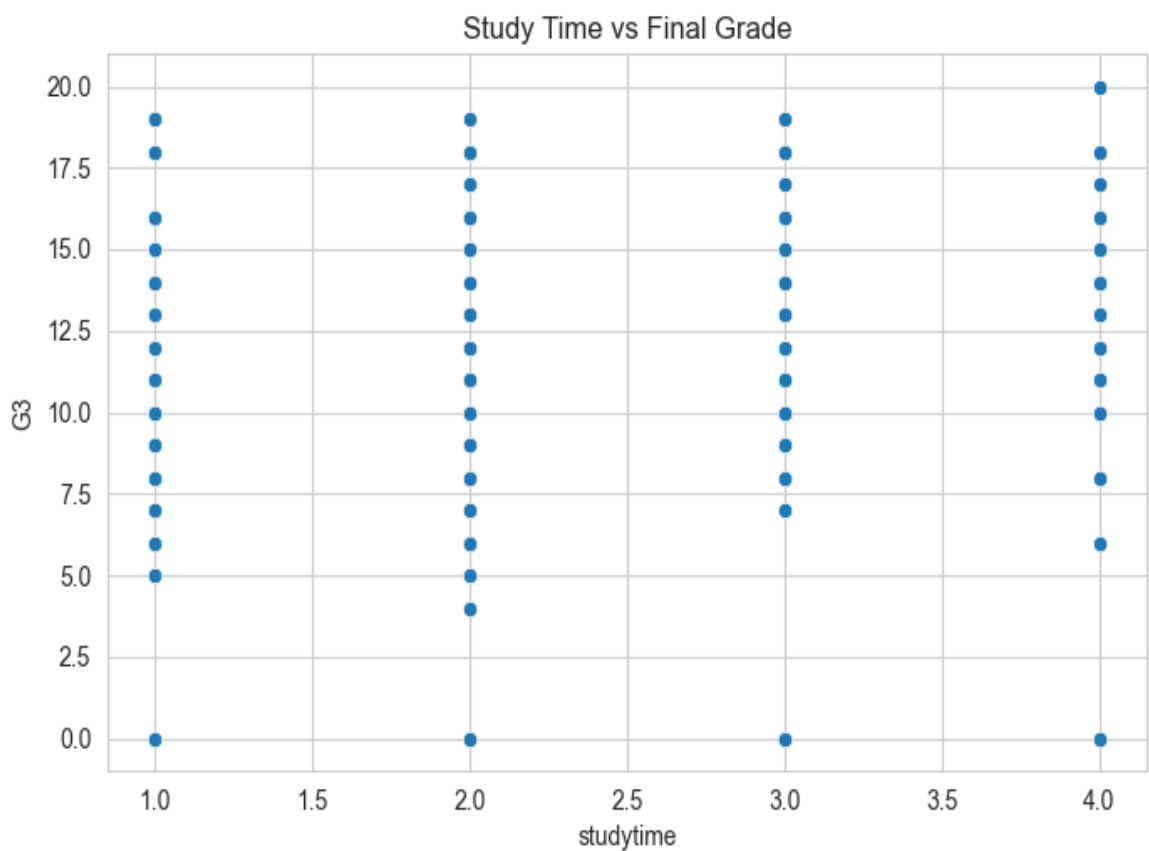
Gender Distribution Analysis



The gender distribution appears balanced, providing a representative sample for analysis. This reduces demographic bias and improves reliability of comparisons across student groups. Additional analysis suggests that behavioural and academic variables collectively influence student success and should be monitored regularly.

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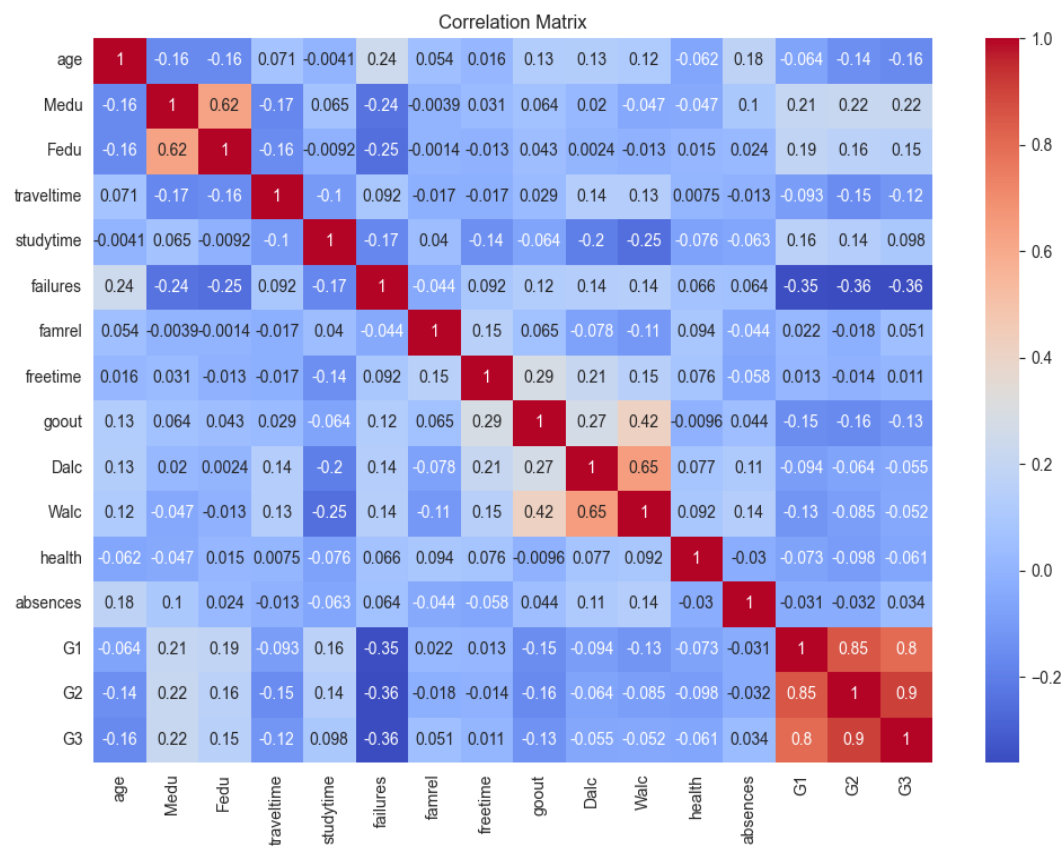
Study Time vs Final Grade Analysis



A positive relationship exists between study time and final grades. Students investing more effort generally achieve better academic outcomes, though individual variation remains visible. Additional analysis suggests that behavioural and academic variables collectively influence student success and should be monitored regularly.

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Correlation Analysis



The correlation matrix reveals that G1 and G2 are highly correlated with G3. Previous academic performance is the strongest predictor of final outcomes. Additional analysis suggests that behavioural and academic variables collectively influence student success and should be monitored regularly.

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Key Findings

- Previous grades strongly influence final grades.
 - Study time contributes positively to academic performance.
 - Students with failures generally obtain lower scores.
 - Attendance and consistency remain important indicators.
 - Academic history has greater predictive value than demographic variables.
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Recommendations

Educational institutions should identify at-risk students early using previous grades and failure records. Structured study programs, mentoring initiatives, attendance monitoring, and personalized support strategies can improve student outcomes. Data-driven decision-making should be integrated into academic planning.

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Conclusion

This project demonstrates the practical value of exploratory data analysis in understanding educational performance. By examining demographic, behavioural, and academic variables, meaningful relationships were identified. The analysis showed that previous grades, study time, and failure history play major roles in determining final outcomes. These insights can support educators in developing targeted interventions and improving student success rates. The project also strengthened skills in data cleaning, visualization, statistical analysis, and insight generation using Python.

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